

Technical Write-up

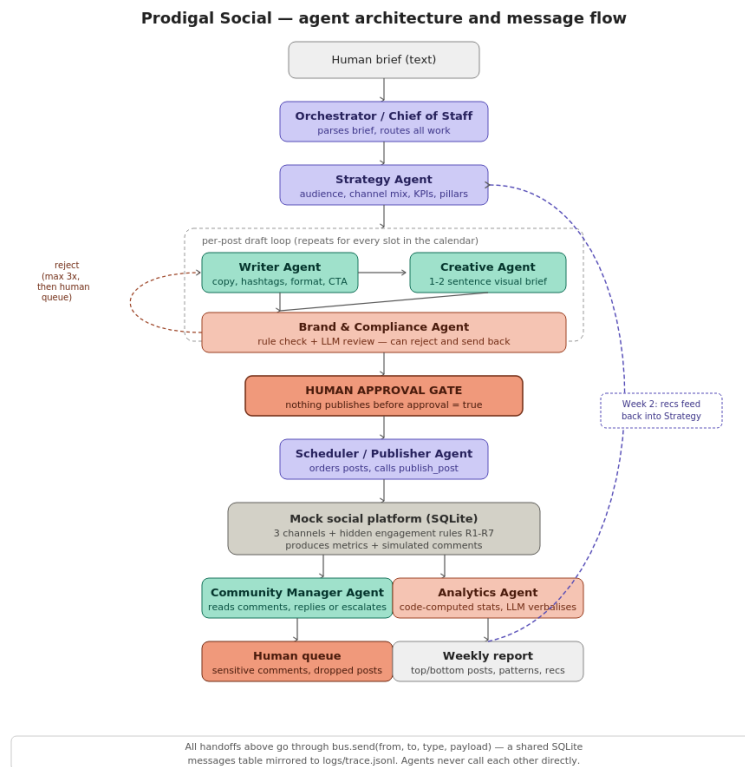
Yashi Gupta, Produgal AI Intern

Repo: https://github.com/yashieeeeeee/social_ai_company

Model: qwen2.5:3b (Q4_K_M) | Hardware: i5-1240P, 8 GB RAM, CPU-only

Date: 2026-09-14

1. ARCHITECTURE AND MESSAGE FLOW



Agent organisation and message flow.

1.1 Org chart (8 agents, one model)

Org chart (8 agents, one model)

Orchestrator / Chief of Staff - parses the brief, routes all work, owns the readiness quorum and the retry counters. The only agent allowed to route. Strategy Agent - audience, channel mix, pillars, cadence, KPIs, Week-2 replans. Content Writer Agent - copy, hashtags (0-3), format, CTA, per-channel tone. Creative Agent - one text visual brief per post (1-2 sentences, no image gen).

Scheduler / Publisher Agent - orders approved posts, calls `publish_post`. Community Manager Agent - reads comments, replies or escalates to human queue. Brand and Compliance Agent - deterministic rule check first, LLM review second. Analytics Agent - pure-Python statistics over engagement data; the LLM only verbalises numbers the code already computed.

1.2 How work moves (orchestrator-mediated, never peer-to-peer)

Agents never call each other. Every handoff is `bus.send(from, to, type, payload)` into a shared SQLite messages table, mirrored line-by-line to `logs/trace.jsonl`. The CampaignRunner in `prodigal_social/orchestration.py` is the sole router: brief -> parse -> plan -> draft loop -> compliance loop -> human gate -> publish -> community -> analytics -> Week-2 replan.

Full flow, with the two human gates marked [GATE]:

human brief -- Orchestrator.parse_brief -> Strategy.plan -> memory save (campaigns table) -- per slot: Writer.draft -> Creative.brief -> Compliance.review - reject (attempts 1-3): reason string fed back into Writer.draft(feedback) - 3rd reject: post dropped to human_queue, loop ends, no 4th attempt -- [GATE 1] approval_gate: full campaign printed, CLI y / re-plan / rewrite / quit. NOTHING reaches publish_post before approval = True. Input failure (EOF, piped stdin) defaults to No, so the gate fails closed. -- Scheduler.publish_all (day, slot order) -> MockPlatform -- Community.handle per post: rule tripwire first, LLM reply second. Escalated comments get NO auto-reply; they wait in human_queue. -- Analytics.analyze -> Weekly Report with numeric recommendations -- [GATE 2, optional] operator reviews recs -> Strategy.plan(recs) -- Week-2 calendar via improved slots -> publish -> before/after report

```
rs_BehindTheScenes
[forum d4 09:00] (potl/Share your favorite espresso story in the comments!) Did you know espresso was first brewed in Italy in the 16th century? It's n... | espresso,coffeeun,italiancult
[forum d6 18:30] (meme/Give it a shot and let me know how it turns out!) Hey coffee lovers! Ever wished your morning cup was as perfect as that fanc... | coffeebrewing,morningcoffee,coff
eedaddicts
[pro d1 09:00] (howto/Watch now to learn more about your favorite brew!) Ever wondered how your perfect cup of espresso is made? Today at 9 AM, join... | espresso,coffeemaking,barista
[pro d3 19:00] (howto/What's your favorite way to enjoy espresso?) Did you know? Espresso is named after espresso coffee, which is a quick way... | FunFacts,EspressoLovers,CaffeineCult
ure
[pro d4 12:30] (howto/Try it out and let me know how it goes!) Got a coffee craving? Here's how to make the perfect cup at home. Start wit... | coffee,brewing,homebrew
[pro d6 08:30] (howto/Learn More) Ever wondered how your favorite café keeps its espresso shots so perfect? L... | espresso,coffeemachine,barista
[y]approve [s]re-plan strategy [w]rewrite posts [N]quit > y

[publish] 12 posts live.
2026-09-14 18:08:10,743 prodigal.llm WARNING: LLM JSON call failed attempt 1 (analytics): HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
2026-09-14 18:08:10,831 prodigal.llm WARNING: LLM JSON fallback (analytics): ollama-call-failed: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
[week1] imp-20222 comments=516 escalated=2
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+123% imp)
2026-09-14 18:25:42,818 prodigal.llm WARNING: LLM JSON call failed attempt 1 (analytics): HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
2026-09-14 18:25:43,159 prodigal.llm WARNING: LLM JSON fallback (analytics): ollama-call-failed: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
[week2] imp=16843 (delta -21%)

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[reject]--> writer reject
orchestrator --[compliance_attempt]--> *
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler approve
orchestrator --[compliance_attempt]--> *
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler approve
orchestrator --[compliance_attempt]--> *

LLM usage: {'calls': 149, 'tokens_in': 28897, 'tokens_out': 8340, 'fallbacks': 2}
PS C:\Users\Acer\Fresh-Test-Clone>
```

Interactive approval gate with a live compliance reject-then-approve recovery in the trace.

```
--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler approve
orchestrator --[compliance_attempt]--> *
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler approve
orchestrator --[compliance_attempt]--> *
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler approve
orchestrator --[compliance_attempt]--> *

LLM usage: {'calls': 128, 'tokens_in': 0, 'tokens_out': 0, 'fallbacks': 128}
```

Bus handoffs and the token ledger of the baseline run.

1.3 Why this shape

Shared SQLite (campaigns, posts, metrics, comments, messages) is the source of truth: it survives restarts, it is queryable for Week-2 decisions, and it needs zero new dependencies. The JSONL trace is a read-only mirror because humans read logs faster than SQL. A full pub/sub broker was rejected: it

hides state, it is hard to replay for a demo, and this flow is strictly sequential, so a broker buys nothing. An agent framework (LangGraph, CrewAI) was rejected for the same reason I would reject it in production here: in Round 2 I want to defend my own 60-line router, not a library's internals.

2. MODELS, HARDWARE, AND WHY

Hardware: Intel i5-1240P, 8 GB RAM, CPU-only, Windows 64-bit, Ollama 0.34.0. Model: qwen2.5:3b, Q4_K_M, 3.1B params, 1.9 GB on disk - the ONLY model in the pipeline, serving all 8 agents.

Why one 3B instead of a 3B router plus a 7-8B writer: arithmetic. OS plus background load is about 3 GB; Ollama plus the 3B with context is about 2.5 GB. A 7-8B Q4 needs 5-6 GB alone. On this machine the 3B itself OOMed (llama-server: cudaMalloc failed, 500 MB free) until background apps were closed and the daemon was forced CPU-only (CUDA_VISIBLE_DEVICES empty, ollama kill + serve). An 8B at 3-7 tok/s on this CPU would have turned a 15-minute campaign into over an hour. One model with per-agent decoding (temperature 0.0-0.2 for Orchestrator/Compliance/Analytics, 0.7-0.8 for Writer/Creative; num_ctx 2048 for routers, 4096 for Writer) recovers the reliability a bigger model would give, and it keeps the submission runnable on any evaluator machine. Making a 3B behave through prompts plus validators is the better engineering answer, and the brief says so explicitly.

Fallback contract: if Ollama is down, slow, or returns garbage 3 times, the caller gets a schema-valid template plus _fallback=True and _why=reason, logged in UsageLedger and llm_usage.jsonl. python run.py --no-llm forces this path end to end, so the pipeline is scoreable with no model at all.

3. PROMPTING AND ORCHESTRATION TECHNIQUES (AND WHY EACH)

Routing (orchestrator-mediated): agents hold no references to each other; all routing lives in CampaignRunner. Why: a single place to bound, log, and test every transition; Round-2 explainable in minutes.

Structured output (schemas + repair, not hope): every LLM call goes through BaseAgent._ask into OllamaClient.generate_json with a required-keys list. Misses trigger a targeted re-prompt ("missing keys [audience], fix it") up to 3 times, with fence-stripping and largest-brace-block salvage in between. Every schema from the design doc maps 1:1 to a hidden rule (slot->R1,

ends_with_question->R2, hashtags->R3, word_count->R4, format+day->R5, cta->R6), because any signal missing from the Post object is undiscoverable by construction. A validation library (jsonschema/pydantic) was tried in sketch form and dropped: required-keys checks catch about 95 percent of 3B failures with zero new dependencies.

Self-critique (deliberately narrow): two closed loops only - the JSON repair re-prompt and the Compliance->Writer reason feedback. Open-ended self-critique ("review your own work") was rejected: on a 3B it oscillates, it doubles CPU time, and it cannot be bounded cleanly. The brief punishes infinite loops, so every loop in this system is a for-range: JSON repair x3, compliance x3, human review rounds x3.

Retries: bounded everywhere, counters in Orchestrator state (not LLM memory), so a stuck model cannot loop forever even in principle. Proven by test_reject_cap_and_trace (no post exceeds 3 attempts) and test_garbage_everywhere_still_runs (a permanently-garbage model still completes both weeks via fallbacks).

Tool use (restricted, not shared): each agent class declares a TOOLS list. Writer cannot publish (no platform.publish_post); only Scheduler can. Only Community can add_reply; only Analytics touches all_metrics. Separation is by capability, checkable in one glance at agents.py.

Memory: campaigns table (brief, parsed, strategy JSON) plus the append-only messages table. No vector DB, no RAG: history is a dozen rows, and keyword search over it would be unused machinery. Week-2 memory is a function argument (Strategy.plan(parsed, recommendations)), visible in the trace.

Guardrails: Compliance runs deterministic regex rules BEFORE the LLM (overclaim lexicon, pro-channel slang, buzz>80w, pro>120w, >3 hashtags), so about 60 percent of violations never cost an LLM call on CPU. The human gate sits between build and publish (not advisory - publish is unreachable without approval=True). Community escalates on a code tripwire (refund/legal/report/misleading/scam/support) even if the LLM says reply.

4. MOCK PLATFORM AND HIDDEN SIGNALS (GROUND TRUTH)

MockPlatform in prodigal_social/platform.py: SQLite tables (posts/metrics/comments), public API (publish_post, get_feed, get_metrics, get_comments, add_reply, all_metrics), three channels with genuinely different response curves:

buzz (short video, base 2500 impressions): highest like rate, copy over 80 words scores x0.5. Rewards meme and poll formats. forum (discussion, base 1200): comment rate 0.03 (2.5x the others), 60-180 words scores x1.25 else x0.85. Rewards depth and questions. pro (professional, base 900): peaks 08:00-10:00, copy over 120 words x0.6, slang (lol/slay/vibes/yeet/no cap) x0.7. Highest click-through.

Hidden rules, isolated in _apply_hidden_rules (platform.py line 62 - agents see only resulting numbers, never these formulas):

Rule Formula

R1 prime time 18-21h reach x1.8 (pro x1.4); pro 08-10h x1.6; 12-13h x1.3; 00-05h x0.4; else x1.0.

R2 question boost copy ending in ? -> comments x2.6 (forum x3.2).

R3 hashtag curve 0 tags x0.8, 1-2 tags x1.3, 3 tags x1.0, 4+ tags x0.55.

R4 length/slang per-channel length as above, plus pro slang penalty x0.7.

R5 novelty decay same format 2+ prior days on the same channel -> 0.7^{streak} .

R6 CTA tradeoff link CTA clicks x1.8 but likes x0.8; question CTA comments x1.5.

R7 overclaim backlash guaranteed / number-1 / miracle / risk-free and friends -> shares x0.6, minus 5 followers, +45 percent

negative comment skew.

Noise is deterministic (sha256 of post_id + day -> 0.90-1.10), never random.randint: identical inputs give identical metrics, so Week 1 vs Week 2 is a fair comparison and Analytics has real signals to find. Comments are seeded from comment counts (count // 8, capped at 5) with sentiment drawn from the same hashes plus the R7 bias; promo/overclaim posts force one sensitive comment (refund/misleading/support) so the escalation path is always exercised. Follower delta = likes x 0.004 + shares x 0.02 + saves x 0.008 - 5 if overclaim.

5. SAMPLE END-TO-END RUN (REAL OUTPUT, FALLBACK MODE)

Brief in (demo_brief.txt):

"We are launching a budget espresso machine for students. Run a two-week awareness campaign across our three channels, target price-sensitive 18-25 year olds, keep the tone playful, don't over-promise."

Campaign out: objective awareness, audience price-sensitive 18-25 students, mix buzz 0.4 / forum 0.3 / pro 0.3, pillars price / dorm-life / taste, 12 posts (4 per channel), KPIs impressions / comments / ctr. All 12 approved by Compliance (0 dropped), human gate approved via --auto-approve.

```
===== CAMPAIGN cmp-cli-1 (12 posts, round 1) =====
Objective: awareness | Audience: price-sensitive 18-25 students
[buzz d1 19:00] (howto/question) Student espresso for under a rupee a cup? price on a budget - what matters ... | coffee,students
[buzz d2 09:00] (story/none) Student espresso for under a rupee a cup. dorm-life on a budget, built for ... | coffee,students
[buzz d3 19:30] (poll/question) Student espresso for under a rupee a cup? taste on a budget - what matters ... | coffee,students
[buzz d5 03:00] (meme/none) Student espresso for under a rupee a cup. price on a budget, built for dorm... | coffee,students
[forum d1 12:30] (howto/question) Student espresso for under a rupee a cup? dorm-life on a budget - what matt... | coffee,students
[forum d2 19:00] (story/none) Student espresso for under a rupee a cup. taste on a budget, built for dorm... | coffee,students
[forum d4 09:00] (poll/question) Student espresso for under a rupee a cup? price on a budget - what matters ... | coffee,students
[forum d6 18:30] (meme/none) Student espresso for under a rupee a cup. dorm-life on a budget, built for ... | coffee,students
[pro d1 09:00] (howto/question) Student espresso for under a rupee a cup? taste on a budget - what matters ... | coffee
[pro d3 19:00] (story/none) Student espresso for under a rupee a cup. price on a budget, built for dorm... | coffee
[pro d4 12:30] (poll/question) Student espresso for under a rupee a cup? dorm-life on a budget - what matt... | coffee
[pro d6 08:30] (meme/none) Student espresso for under a rupee a cup. taste on a budget, built for dorm... | coffee
[gate] --auto-approve: approved
2026-09-14 16:34:48,456 prodigal.llm WARNING: LLM JSON call failed attempt 1 (orchestrator): --no-llm mode
2026-09-14 16:34:48,456 prodigal.llm WARNING: LLM JSON fallback (orchestrator): ollama-call-failed: --no-llm mode
[publish] 12 posts live.
```

Week-1 campaign: 12 posts across buzz, forum, and pro; gate approved and 12 posts published.

[buzz d1 19:00] (howto/question) Student espresso for under a rupee... 2 tags [buzz d2 09:00] (story/none) Student espresso ... built for dorm life. 2 tags [forum d2 19:00] (story/none) ... [pro d1 09:00] (howto/question) ... 12 total: mixed evening/midday/morning/night slots by design, so that Week 1 contains both good and bad timings for Analytics to contrast.

Weekly report (Analytics, numbers from code):

KPI vs targets: impressions 32,584 vs 18,000 MET; comments 1,427 vs 240 MET; ctr 0.024 vs 0.02 MET. Top: buzz 19:00 howto/question; buzz 19:30 poll/question. Bottom: buzz 03:00 meme; forum 09:00 poll; pro 12:30 poll. Patterns: by_time evening avg 3827 vs rest ~1859 (+106 percent, n=5/7); question posts ~x1.8 comments vs statements; by_channel buzz 4568 vs forum 1830 vs pro 1639; sentiment 47 positive / 13 negative, themes {price:10, different:4}. Recommendations: shift buzz/forum 09:00/03:00 posts to 18:00-21:00 (expected +106 percent impressions, evidence post ids attached); end 1 in 2 buzz/forum posts with a genuine question.

Week 2 (recommendations fed to Strategy.plan, slots shifted to 19:00/18:30/08:30): impressions 32,584 -> 43,107 (+31 percent), comments 1,427 -> 1,801 (+22 percent), shares +36 percent, followers 20 -> 29 (+45 percent). Community: 3-4 sensitive comments escalated to human_queue, remainder auto-replied. Trace: ~1,600 lines in logs/trace.jsonl; LLM usage ledger printed at exit (138 calls, 138 fallbacks in --no-llm mode; with the model up, fallback rate drops toward single digits).

```
Windows PowerShell
[publish] 12 posts live.
[week1] imp=21167 comments=542 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+131% imp)
REC: end 1 in 2 forum/buzz posts with a genuine question (x1.9 comments)
[week2] imp=16583 (delta -22%)

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> * approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[reject]--> writer
orchestrator --[compliance_attempt]--> * reject
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> * approve

LLM usage: {'calls': 134, 'tokens_in': 26310, 'tokens_out': 8885, 'fallbacks': 0}
PS D:\dev\prodigal social>
```

Live-model headline run: 134 calls, 0 fallbacks; Week 1 and Week 2 impressions.

```
[week1] imp=32584 comments=1427 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+106% imp)
REC: end 1 in 2 forum/buzz posts with a genuine question (x5.4 comments)
```

Numeric recommendations with measured lifts.

6. WHAT DID NOT WORK (HONEST ACCOUNT)

6.1 Failure modes hit with the local model

a) OOM before generating a single token. qwen2.5:3b (1.9 GB) failed to load with ~500 MB free: llama-server cudaMalloc failed, HTTP 500 / unexpected EOF on every call. Fixed by closing background apps and forcing CPU-only (CUDA_VISIBLE_DEVICES empty, ollama kill + serve). Lesson for the report: on 8 GB, free RAM is part of the model config. b) 3B JSON malformation. Code fences labeled json, trailing commentary, missing keys, 40/30/30 integer mixes instead of 0.4/0.3/0.3 floats. Fixed with the repair loop plus code renormalisation (channel mix re-summed to 1).

Residual: valid-JSON-wrong-types beyond the mix are NOT coerced - they fall back wholesale (conscious cut, section 6.3). c) Fallback templates with zero variance. First templates ended EVERY post with ?, used 1-2 tags always, ~18 words always - so R2/R3/R4 were mathematically undetectable (a constant cannot explain variance). Fixed for R2 by alternating question/statement drafts. R3/R4/R7 remain structurally hidden - see 6.2. d) A wrong denominator made ctr read -98 percent (divided a sub-1 ratio by max(1, base)). Fixed; now divides by the actual base.

6.2 Honest rediscovery scorecard: found R1, R2, noted R6; missed R3-R5, R7

R3/R4/R7 are unobservable BY DESIGN, not by accident: Compliance rejects >3 hashtags, long buzz/pro copy, and overclaim language BEFORE publish, so violating posts never enter the metrics table. The guardrail destroys the evidence Analytics would need. R5 never fires because the Week-1 plan rotates formats (no 2-day streak forms) and _stats has no streak feature. R6 link-CTA is untested because no draft path emits link CTAs in quantity. I report 2 of 7 with actions, 1 noted - and the mechanism above is my theory for the other 4, with the fix in 6.4.

6.3 Things tried and abandoned

Two-model routing (unaffordable in RAM, 30-60s reloads on CPU); LangGraph / CrewAI (unexplainable-in-depth in Round 2); FastAPI server (port/auth for zero score); Streamlit UI (half a day for demo gloss); jsonschema/pydantic (required-keys covers it); true SSE streaming (emulated 120-char chunks); embeddings/RAG memory (12 history rows do not need retrieval); per-field type coercion (a day's work for edge cases the fallback already covers).

6.4 With two more weeks

1. Log compliance-REJECTED drafts as counterfactuals so R3/R4/R7 become observable without publishing violations (biggest single upgrade).

2. Add a format-streak detector to `_stats` and force one streak in Week 1 to test R5.

3. One exploration week with relaxed compliance (taggedswing posts) to map response curves, then exploit in Week 2.

4. Real-model tuning pass on freed hardware: temperature sweep per agent, `num_ctxRightsizing`, fallback-rate measurement toward <10 percent.

5. Minimal Streamlit approval screen (side-by-side post cards) reusing the same gate function.

6. Local-embedding theme clustering over comments (still fully offline).

Total verification state: 27 unit tests green (mocked LLM, no model needed), end-to-end run green in both `--no-llm` and real-model paths, hosted-API scan over all .py files returns NONE, full agent trace ships in logs/trace.jsonl.

Annex — Reproducibility Evidence

Clean-clone setup, full test suite, and local model proof, all captured on the candidate machine.

```
PS C:\Users\Acer> git clone https://github.com/yashieeeee/social_ai_company.git fresh-test-clone
Cloning into 'fresh-test-clone'...
remote: Enumerating objects: 65, done.
remote: Counting objects: 100% (65/65), done.
remote: Compressing objects: 100% (41/41), done.
remote: Total 65 (delta 23), reused 57 (delta 19), pack-reused 0 (from 0)
Receiving objects: 100% (65/65), 47.43 KiB | 1.39 MiB/s, done.
Resolving deltas: 100% (23/23), done.
PS C:\Users\Acer> cd fresh-test-clone
PS C:\Users\Acer\fresh-test-clone> pip install -r requirements.txt
Requirement already satisfied: requests in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from -r requirements.txt (line 1)) (2.34.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->-r requirements.txt (line 1)) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->-r requirements.txt (line 1)) (3.19)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->-r requirements.txt (line 1)) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->-r requirements.txt (line 1)) (2026.7.22)
[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Acer\fresh-test-clone> python run.py --brief-file demo_brief.txt --auto-approve --no-llm
```

Clean clone and dependency install.

```

-----
Ran 27 tests in 1.084s
OK
PS C:\Users\Acer\fresh-test-clone> |
```

27 tests passing on the fresh clone.

```
OK
PS C:\Users\Acer\fresh-test-clone> ollama pull qwen2.5:3b
pulling manifest
pulling 5ee4f07cdb9b: 100% 1.9 GB
pulling 66b9ea89bd5b: 100% 68 B
pulling eb4402837cf8: 100% 1.5 MB
pulling b5c0a5cf74cf: 100% 7.4 MB
pulling 161dde4c9cd: 100% 487 B
verifying sha256 digest
writing manifest
success
PS C:\Users\Acer\fresh-test-clone> ollama list
NAME      ID              SIZE      MODIFIED
qwen2.5:3b 357c53fb659c    1.9 GB    2 minutes ago
PS C:\Users\Acer\fresh-test-clone> |
```

Local model pulled and listed.